



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Software Engineering

CS-405 Deep Learning

BESE-12

Lab 7: Open Ended Lab

Date: 01-11-2024

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Introduction

This lab is designed to provide students with hands-on experience in applying transfer learning techniques to the field of plant disease detection. The open-ended nature of this lab encourages students to explore solutions independently, fostering creativity and deeper understanding.

Objectives

By the end of this lab, students should be able to:

- Apply transfer learning methods without relying on step-by-step guidance.
- Analyze and interpret results from transfer learning models.

Tools

Google Colab

Lab Task

For this lab, you will use the plant disease dataset available on Kaggle. Please download the dataset from the following link:

<https://www.kaggle.com/datasets/vipooooool/new-plant-diseases-dataset>

Your tasks are as follows:

1. Select two Convolutional Neural Network (CNN) architectures from different families.
2. Apply transfer learning on the selected architectures by:
 - a. Fine-tuning the model.
 - b. Using the model as a feature extractor.
3. Compare and analyze the results of both approaches.

Submission:

Each student is required to submit the following:

- A Jupyter notebook that includes:
 - A detailed explanation of the dataset.
 - A description of the pipeline and methods used.
 - An analysis of the results from each approach.